



Title

rdbwhite — Data-driven bandwidth selection for RD Heterogeneous Treatment Effects.

Syntax

```
rdbwhite depvar runvar [if] [in] [, covs_hte(covars) c(#) p(#) q(#)
kernel(kernelfn) weights(wtvar) vce(vcetype [vceopt1 vceopt2])
covs_eff(covars) bwselect(bwmethod) precision(single|double) bwjoint labels ]
```

Description

rdbwhite provides data-driven bandwidth selection for estimation and inference of heterogeneous treatment effects in RD designs ([Calonico, Cattaneo, Farrell, Palomba and Titiunik, 2025a](#)).

Companion commands are: **rdhte** for estimation and inference, and **rdhte_lincom** for testing linear restrictions of parameters.

A detailed introduction to **rdhte** in Stata is given in [Calonico, Cattaneo, Farrell, Palomba and Titiunik \(2025b\)](#).

Related software packages for analysis and interpretation of RD designs and related methods are available in:

<https://rdpackages.github.io/>

For background methodology, see [Calonico, Cattaneo, Farrell, and Titiunik \(2019\)](#), [Calonico, Cattaneo and Farrell \(2020\)](#), [Cattaneo and Titiunik \(2022\)](#).

Options

Estimand

c(#) specifies the RD cutoff for *indepvar*. Default is **c(0)**.

covs_hte(covars) specifies covariate(s) for heterogeneous treatment effects. Factor variables notation can be used to distinguish between continuous and categorical variables, select reference categories, specify interactions between variables, and include polynomials of continuous variables. If not specified, the RD Average Treatment Effect bandwidth is computed.

labels displays the final bandwidth estimates using variable labels from **covs_hte**(covars).

Local Polynomial Regression

p(#) specifies the order of the local polynomial used to construct the point estimator. Default is **p(1)** (local linear regression).

q(#) specifies the order of the local polynomial used to construct the bias correction. Default is **q(2)** (local quadratic regression).

kernel(kernelfn) specifies the kernel function used to construct the local-polynomial estimator(s). Options are: **triangular**, **epanechnikov**, and **uniform**. Default is **kernel(triangular)**.

weights(wtvar) specifies a non-negative variable used to weight the bandwidth-selection procedure. The unit-specific weights multiply the kernel function.

covs_eff(covars) specifies additional covariates to be used for efficiency improvements.

precision(*single|double*) controls the storage type used for generated working variables, including the centered running variable and treatment indicator. The default is **precision(double)**. Use **precision(single)** to restore Stata's legacy float temporary-variable path for backward numerical compatibility.

Data-Driven Bandwidth Selection

bwselect(*bwmmethod*) specifies the bandwidth selection procedure to be used.

Options are:

mserd one common MSE-optimal bandwidth selector for the RD treatment effect estimator.

msetwo two different MSE-optimal bandwidth selectors (below and above the cutoff).

msesum one common MSE-optimal bandwidth selector for the sum of regression estimates.

msecomb1, **msecomb2**, **cerrd**, **certwo**, **cersum**, **cercomb1**, **cercomb2**: see [rdhte](#).

Default is **bwselect(mserd)**.

bwjoint forces all bandwidths to be the same across groups. By default, separate bandwidths are computed for each group when **covs_hte()** is a factor or 0/1 variable; for continuous **covs_hte()** a single joint bandwidth is always used regardless of **bwjoint**.

Variance-Covariance Estimation

vce(*vcetype* [*vceopt1 vceopt2*]) specifies the procedure used to compute the variance-covariance matrix estimator. Options are:

vce(hc0) heteroskedasticity-robust plug-in residuals, no weights.

vce(hc1) heteroskedasticity-robust plug-in residuals, HC1 weights.

vce(hc2) heteroskedasticity-robust plug-in residuals, HC2 weights (leverage).

vce(hc3) heteroskedasticity-robust plug-in residuals, HC3 weights (jackknife).

vce(cr1 clustervar) cluster-robust CR1 variance with $((n-1)/(n-k)) * (G/(G-1))$ multiplier. Default when a cluster variable is supplied.

vce(cr2 clustervar) cluster-robust CR2 variance (Bell-McCaffrey block-adjusted residuals).

vce(cr3 clustervar) cluster-robust CR3 variance (block jackknife, $(G-1)/G$ multiplier).

vce(cluster clustervar) alias for **vce(cr1 clustervar)**.

Legacy: **vce(hc0/hc1 clustervar)** is remapped to **cr1**, **vce(hc2 clustervar)** to **cr2**, **vce(hc3 clustervar)** to **cr3** (with a warning).

Default is **vce(hc3)** when no cluster, **vce(cr1 clustervar)** when a cluster variable is supplied.

Example:

Setup using [Granzier, Pons, and Tricaud \(2023\)](#) Data

```
. use rdrobust_senate.dta
```

RD-HTE Estimation by left/right groups

```
. rdbwhite y x, covs_hte(i.left)
```

RD-HTE Estimation using a continuous variable

```
. rdbwhite y x, covs_hte(c.mean_strength_nat1)
```

Stored results

rdbwhite stores the following in **e()**:

Scalars

e(N) original number of observations
e(c) cutoff value
e(p) order of the polynomial used for the point estimator
e(q) order of the polynomial used for bias correction

Macros

e(cmd) **rdbwhite**
e(depvar) name of dependent variable
e(outcomevar) name of outcome variable (alias for **e(depvar)**)
e(runningvar) name of running variable
e(clustvar) name of cluster variable, when **vce(cluster ...)** or **vce(hc2 ...)** was specified
e(covs) names of efficiency covariates from **covs_eff()**, when supplied
e(vce_select) vcetype label
e(kernel) kernel type used
e(bwselect) bandwidth selection procedure
e(precision) storage precision used for generated working variables

Matrices

e(h) I x 2 matrix of left/right bandwidths, one row per group
e(tau_N) I x 2 effective sample sizes per group, per side

References

- Calonico, Cattaneo, Farrell, Palomba and Titiunik. 2025a. Treatment Effect Heterogeneity in Regression Discontinuity Designs. *Working Paper*.
- Calonico, Cattaneo, Farrell, Palomba and Titiunik. 2025b. rdhte: Conditional Average Treatment Effects in RD Designs. *Working Paper*.
- Granzier, Pons, and Tricaud. 2023. Coordination and Bandwagon Effects: How Past Rankings Shape the Behavior of Voters and Candidates. *American Economic Journal: Applied Economics*, 15(4): 177-217.
- Cattaneo and Titiunik. 2022. Regression Discontinuity Designs. *Annual Review of Economics*, 14: 821-851.
- Calonico, S., M. D. Cattaneo, and M. H. Farrell. 2020. Optimal Bandwidth Choice for Robust Bias Corrected Inference in Regression Discontinuity Designs. *Econometrics Journal*, 23(2): 192-210.
- Calonico, Cattaneo, Farrell, and Titiunik. 2019. Regression Discontinuity Designs using Covariates. *Review of Economics and Statistics*, 101(3): 442-451.
- Calonico, Cattaneo, and Titiunik. 2014. Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs. *Econometrica*, 82(6): 2295-2326.

Authors

Sebastian Calonico, University of California, Davis, CA. scalonico@ucdavis.edu.

Matias D. Cattaneo, Princeton University, Princeton, NJ.
matias.d.cattaneo@gmail.com.

Max H. Farrell, University of California, Santa Barbara, CA. mhfarrell@gmail.com.

Filippo Palomba, Princeton University, Princeton, NJ.
filippo.palomba19@gmail.com.

Rocio Titiunik, Princeton University, Princeton, NJ. rocio.titiunik@gmail.com.